Do national health and wealth predict attitudes toward, and use of mouth-to-mouth kissing in romantic relationships?

CHRISTOPHER D WATKINS[1*+], JUAN DAVID LEONGÓMEZ[2+], JEANNE BOVET[3+], AGNIESZKA ZELAZNIEWICZ[4], MAX KORBMACHER[1], MARCO ANTONIO CORREA VARELLA[5], ANNA MARIA FERNANDEZ[6], DANIELLE WAGSTAFF[7] & SAMUELA BOLGAN[1]

[1]Division of Psychology, School of Social and Health Sciences, Abertay University, Dundee, Scotland DD11HG

[2]Facultad de Psicologia, Universidad El Bosque, Bogota, Colombia

[3]Institute for Advanced Study in Toulouse, Toulouse, France

[4]Universytet Wroclawski, Wroclaw, Poland

[5]Department of Experimental Psychology, University of Sao Paulo, Sao Paulo, Brazil

[6]USACH, Escuela de Psicologia, University of Santiago, Chile

[7]Faculty of Health, Federation University Australia, Balarat, Victoria, Australia

+ Denotes authors who commented on the pre-registration report. Report compiled after data collection but before data analysis.

***Corresponding author:**

Dr Christopher Watkins

Division of Psychology, School of Social and Health Sciences,

Abertay University, Dundee, Scotland DD11HG

Tel: (44)1382 308646

Introduction

Romantic love and passion are cultural universals (Jankowiak, 1995, 2008; Jankowiak & Fischer, 1992; Jankowiak & Paladino, 2008). Simultaneously, pair bonds within different cultures vary in their norms, rituals and forms of romantic expression (Hatfield & Rapson, 2005). In western samples, 'ideal relationships' are conceived on two dimensions of intimacy-loyalty and passion (Fletcher et al., 1999), with intimacy related to relationship quality independent of couple sexuality (Hassebrauck & Fehr, 2002). Expressions of love and overt acts of affection enhance feelings of commitment (Marston et al., 1998) and are related to stable marital bonds (Huston et al., 2001). Moreover, sharing in novel and arousing activities enhances relationship quality (Aron et al., 2000) and nonverbal expressions of love alter feeling states and facilitate the release of putative attachment hormones (Gonzaga et al., 2006). Collectively, many relationship behaviours and expressions of romantic attachment are important in maintaining durable pair bonds, which are important to understand given that threats to pair bonds may effect health and wellbeing (see Cacioppo et al., 2015 for discussion).

Although not a human universal, romantic kissing is observed in a wide variety of cultures (Jankowiak et al., 2015) and being perceived as a good kisser can enhance a person's desirability as a partner (e.g., for short-term relationships, Wlodarski & Dunbar, 2014). Indeed, some naturalistic studies have documented kissing in courtship, where reciprocal affection is related to synchrony in body movements (Brak-Lamy, 2015). Cues obtained from close contact with a partner may facilitate mate assessment during courtship (Wlodarski & Dunbar, 2013a), consistent with species who use gustatory and olfactory pheromones to regulate courtship rituals (Clowney et al., 2015; see also Thistle et al., 2012). As the prevalence of courtship rituals may point to their adaptive function, one hypothesis (the mate assessment hypothesis; Wlodarski & Dunbar, 2013a, 2013b) proposes that kissing functions to assess putative cues to biological quality in a partner via close contact. In light of behavioural motivations to avoid pathogens (see Tybur & Lieberman, 2016 for discussion), biases to over-perceive disease cues (Miller & Maner, 2012), and given that the exchange of saliva may increase the likelihood of transmitting some infections (see Posse et al., 2017 for a recent review), romantic kissing may incur costs. As such, we do not engage in romantic kissing indiscriminately, unless the costs of kissing are traded off in favour of escalating courtship with a 'high quality' mate (Wlodarski & Dunbar, 2013a). Consistent with the mate assessment hypothesis, kissing is more important, and is more likely to influence attraction, for the more 'selective' sex (women, Todd et al., 2007; Trivers, 1972), among more selective individuals (attractive individuals) and in contexts where the costs of choosing a less healthy mate are greater (i.e. in short-term sexual encounters; Wlodarski & Dunbar, 2013a). The pair bonding hypothesis posits that, as an expression of love that strengthens romantic attachment, kissing plays a functional role in monitoring and maintaining long-term relationship quality (Wlodarski & Dunbar, 2013a). Consistent with this hypothesis, kissing frequency and satisfaction with kissing are related to relationship quality while having a partner who is a good kisser predicts relationship quality at twice the size of the relationship between sexual satisfaction and relationship quality (Wlodarski & Dunbar, 2013b). Collectively, kissing may be important both as a courtship custom and in maintaining stable long-term pair bonds.

Here, we extend Wlodarski & Dunbar's (2013a, 2013b) two hypotheses, and test for cultural differences in the use and appreciation of kissing in romantic relationships. First, if kissing plays a role in assessments of ‘quality’, the benefits of assessing partner quality are likely to be greater in less-healthy environments (see, e.g., Gangestad et al., 2006 for a similar line of reasoning related to mate preferences). Here, we examine the mate assessment hypothesis and test whether people in less-healthy countries place greater importance on this form of courtship behaviour (i.e. to assess quality, as indexed via attitudes) but do so selectively (as indexed via frequency) in light of the greater potential risks of kissing within a high-pathogen environment. Thus, we predict that national health will be negatively related to the importance of kissing at the *initial* stages of a romantic relationship (Hypothesis #1) and will be negatively related to participant’s reported satisfaction with the *amount* of kissing in a romantic relationship (Hypothesis #2). Moreover, we predict that national health will be positively related to the frequency of kissing within a romantic relationship (Hypothesis #3). .

Second, theoretical perspectives argue that monogamy and/or relationship investment are valued in harsher environments, such as those where resources are scarce (see, e.g., Gangestad & Simpson, 2000; Little et al., 2011 for discussion), and the pair bonding hypothesis (Wlodarski & Dunbar, 2013a) proposes that kissing plays an important role in how couples maintain and monitor the quality of a committed romantic relationship. Thus, we test the pair bonding hypothesis by examining whether individuals from countries of low absolute and relative wealth (i.e. high income inequality) place greater importance on kissing at the *established* phases of a romantic relationship (Hypothesis #4), report greater frequency of kissing (Hypothesis #5) and lower satisfaction with the amount of kissing in their relationships (Hypothesis #6).

When testing both hypotheses on the function of mouth-to-mouth kissing in relationships, we also test for national differences in two other forms of close/intimate contact (hugging and sexual intercourse) in order to examine whether our predictions are specific to, or are stronger for, mouth-to-mouth kissing than other forms of romantic expression. Evidence that this is the case would extend prior work, which demonstrates that kissing is a more substantial predictor of romantic relationship quality than other forms of closeness and/or intimacy such as sexual intercourse (Wlodarski & Dunbar, 2013b), by suggesting that the 'special' role that kissing may play within longer-term romantic relationships could vary according to the harshness of the environment.

Finally, we aim to replicate prior work by testing for an identical factor structure to the perceived components of a good kiss among a diverse international sample (biological component, arousal/contact and technique/execution; Wlodarski & Dunbar, 2013b). Here, we test whether participant's factor scores on the biological component of a good kiss are negatively related to national health, which would demonstrate that biological factors (e.g., related to gustatory and olfactory cues) are a stronger component of a 'good kiss' in environments where exposure to pathogens is of greater concern (Hypothesis #7). We will also run exploratory analyses on the other two factors (arousal/contact and technique/execution) and measures of national wealth to test whether these factors, which suggest greater importance attached to synchrony, arousal and bonding during kissing, are valued more in environments where cues to an investing partner are likely to be at a premium.

Methods

Participants

A large (>3000) online adult sample was recruited to take part in an online study on romantic expression across cultures, with no exclusion criteria set for participation. Data collection commenced in February 2017 and ended after collating data from 13 countries (6 continents). Countries are included in cross-national comparisons if we have data from at least 50 participants (respondents who reside in the same country as their birth) and responses to our kissing questionnaire from at least 30 females from that same country. Participants were a sample of convenience from campuses and the wider community, recruited via research participant pools, word of mouth via colleagues, twitter, and posts to academic groups on social media and a press release from the lead author's communications department to coincide with Valentine's Day 2017. This latter press release informed the public that we were conducting a global study into how people express themselves in romantic relationships, but did not mention our predictor variables of national health or national wealth. Given difficulty in recruiting an African sample, we recruited a sample of Nigerians via the buy responses function on surveymonkey.com. These participants were not reimbursed for their time (the platform provides donations to charity in exchange for participation).

All procedures for testing and recruitment were approved via the lead author's local Ethics Committee. Participants provided informed consent after reading an information sheet describing the contents of the survey. We excluded participants who i) reported being less than 18 years old, ii) did not report their sex as male or female, or, for analyses involving cross-national comparisons, iii) if their IP address did not match their reported country of residence (following DeBruine et al., 2010).

Measures

Participants first provided demographic information (sex, age, sexual orientation, country of residence, country of birth, current involvement in a long-term romantic relationship, relationship length, ethnicity, self-rated attractiveness, self-rated masculinity, self-rated health, rated attractiveness/masculinity/health of current romantic partner). Participants then completed three questionnaires in a randomized order. Two of these questionnaires (confessing love to partner and attachment anxiety) were unrelated to the current study and were run as part of a separate project.

In the current study, participants completed a questionnaire about their attitudes toward mouth-to-mouth kissing, adapted from Wlodarski & Dunbar (2013a, 2013b). Participants were informed that we were interested in their views on various behaviours that a person may engage in with a romantic partner, whom we defined as a person with whom you could be romantically involved with or without the involvement of sex. Kissing was defined as kissing on the lips or open-mouth (i.e. ‘French’ kissing). Participants were asked how important they thought kissing was i) at the very initial stages of a romantic relationship and ii) during the established phases of a committed, long-term relationship on a 0 (not at all important) to 100 (extremely important) scale, with their choice on the scale visible. Participants were asked, in general, when they are in a romantic relationship i) how often they kiss their partner, ii) just hug or cuddle their partner (a single hug or cuddle without kissing involved), iii) have sexual intercourse with their partner, on a 0 (not at all) to 100 (very often) scale. Participants were asked to indicate their satisfaction with the amount of kissing/hugging-cuddling/sexual intercourse (in general, when they are in a romantic relationship) on a zero (not at all satisfied) to 100 (very satisfied) scale.

Finally, following Wlodarski & Dunbar (2013b) participants were asked to indicate the importance of the following factors when deciding whether someone is a ‘good kisser’ on a 0 (not at all important) to 100 (extremely important) scale i) How pleasant their breath is, ii) The scent of their body, iii) The taste of their lips/skin, iv) How ‘wet’ the kiss is, v) How much touching/physical-contact/caressing is involved, vi) How physically aroused it makes you, vii) Whether their kissing style is the same as yours. After completing all questionnaires, participants were debriefed and could exit the survey. Foreign language versions of the study were available (French, Spanish, Portuguese, Italian, German, Polish), and had been translated by native speakers based at a university.

Following Tybur et al. (2016) we measured national differences in parasite stress using data on the historical prevalence of pathogens within regions (9-item measure, Murray & Schaller, 2010) which is strongly correlated with estimates of parasite stress derived from other instruments and provides converging evidence when the same hypotheses are tested using alternate measures of parasite stress (see Tybur et al., 2016). Following other cross-cultural research (e.g., Brooks et al., 2011; Daly et al., 2001), national differences in absolute wealth and income inequality were measured via national GDP per capita and the GINI Index (inequality in the distribution of family income) obtained from the CIA World Factbook (https://www.cia.gov/library/publications/the-world-factbook/). High scores indicate greater absolute wealth and greater income inequality respectively.

Analytical strategy

Initial analyses. First, we will run general linear models to examine kissing behaviour and attitudes across the entire sample to see if we replicate prior findings (Wlodarski & Dunbar, 2013a) with a larger global sample (i.e. individual differences in attitudes toward kissing in different relationship contexts, as a function of the sex of participant and their self-rated attractiveness). Analyses on data from the entire sample will not exclude participants according to whether their country of birth matches their current country of residence, in contrast to analyses where we test for cultural differences. We apply this latter control as a conservative measure given that this is an online cross-cultural study where we do not collect data on residence duration.

Cultural differences in attitudes and behaviour related to kissing. Data will be analysed via linear mixed models with maximum likelihood estimation criteria. Bias-corrected bootstrapped confidence intervals will be reported for all cross-cultural analyses. Following prior recommendations (Pollet et al., 2017), participants (level 1 units) will be nested within nations (level 2 units) and nations will be treated as random effects. Following prior cross-cultural research (Tybur et al., 2016), and evidence for sex differences in response to our questions (Wlodarski & Dunbar, 2013a), we will control for participant sex by including this variable in our model as a fixed effect. Given that we anticipate skews in relationship status within the sample and differences between nations in the mean age of the sample, we will also include age and relationship status as fixed effects. Participant sex will not be entered as an interaction term in our model given that we anticipate not having sufficient power to analyse sex as a moderator of the hypothesised links between our predictor and outcome variables (i.e. due to few male respondents in some of our national samples). Self-rated attractiveness will be entered as a control variable in light of prior findings discussed earlier, where this variable moderates attitudes toward kissing (Wlodarski & Dunbar, 2013a).

Our full model, for hypothesis # 1 (national health negatively related to importance of kissing at the initial stages of a relationship) is therefore as follows:

$$\begin{aligned}
&\boldsymbol{\gamma}_{\boldsymbol{Kiss\ Importance\ Initial}} \\
&\quad = (\boldsymbol{b_0} + \boldsymbol{u_{0,Nation}}) + \boldsymbol{b_1 National\ Historical\ Parasite\ Stress_i} \\
&\quad + \boldsymbol{b_2 National\ GINI\ Coefficient_i} + \boldsymbol{b_3 National\ Wealth}\ (\boldsymbol{absolute})_{\boldsymbol{i}} \\
&\quad + \boldsymbol{b_4 Relationship\ status_i} + \boldsymbol{b_5 Participant\ sex_i} \\
&\quad + \boldsymbol{b_6 Participant\ age_i} + \boldsymbol{b_7 Self\ rated\ attractiveness_i} + \boldsymbol{\varepsilon_i}
\end{aligned}$$

To assess any unique effects of the ecological predictor variables (National Historical parasite stress, GINI coefficient and Wealth) on our outcome variable, we compare a first model (including national historical parasite stress only) against a null model using likelihood ratio tests and then add each ecological variable incrementally to our model to test whether the predictive power of the model improves. Although our three ecological variables may have problems with multicollinearity (but see Brooks et al., 2011), inclusion of these variables are theoretically driven and used to directly test the two hypotheses on the functions of kissing in romantic relationships (mate assessment versus pair bonding). Thus, an alternate strategy, where ecological variables are subject first to principle components analysis (Kandrick et al., 2015) would not enable us to test our hypotheses in this instance (e.g., if the resulting analysis loaded onto one primary factor of ‘harshness’).

This process will then be repeated for our remaining outcome variables (behaviours and/or attitudes related to kissing, hugging and sexual activity) in order to test hypotheses 2-6 and examine whether outcome variables vary across nations as a function of our level two ecological variables. Following Tybur et al. (2016), relationships will be illustrated via scatterplots showing correlations between outcome variables aggregated at the national level and our ecological predictor variables, after controlling for sample demographic characteristics.

Cultural differences in the components of a 'good kiss'. For analyses testing for relationships between factor scores derived from responses to the 'qualities of a good kiss' questionnaire, we will test whether individual participant's scores on each factor (level 1 variable) are correlated with our level 2 ecological variables. Factor scores will be generated via factor analysis of responses to these questions across the full sample. Assuming this process generates identical factors to Wlodarski & Dunbar (2013b), we will then examine whether these factor scores vary across nations as a function of our level 2 ecological variables, using likelihood ratio tests to assess the predictive power of our model against a null model.

For example, to assess whether biological factors are a stronger component to a 'good kiss' in less healthy countries (Hypothesis #7):

$$\begin{aligned}\gamma_{Biological\ Factor-Good\ kiss} &= (\boldsymbol{b_0} + \boldsymbol{u_{0,Nation}}) + \boldsymbol{b_1 National\ Historical\ Parasite\ Stress_i} \\ &+ \boldsymbol{b_2 Partnership\ status_i} + \boldsymbol{b_3 Participant\ sex_i} \\ &+ \boldsymbol{b_4 Participant\ age_i} + \boldsymbol{\varepsilon_i}\end{aligned}$$

Finally, in light of concerns that clustering of traits around broader geopolitical regions may partly account for observed differences in behaviour at the national level (Pollet et al., 2017), alternate analyses will be run to confirm that any positive cross-cultural observed are not artefacts of the national samples that we have access to. We will account for this by including United Nations regional group (dummy coded) as a level three variable in further analyses of positive findings.

References

Aron, A., Norman, C. C., Aron, E. N. & McKenna, C. (2000). Couples' shared participation in novel and arousing activities and experienced relationship quality. *Journal of Personality and Social Psychology, 78*, 273-284

Brak-Lamy, G. (2015). Heterosexual Seduction in the Urban Night Context: Behaviors and Meanings. *Journal of Sex Research, 52*, 690-699.

Brooks, R., Scott, I. M., Maklakov, A. A., Kasumovic, M. M., Clark, A. P. & Penton-Voak, I. S. (2011). National income inequality predicts women's preferences for masculinized faces better than health does. *Proceedings of the Royal Society of London B, 278*, 810-812.

Cacioppo, J. T., Cacioppo, S., Capitanio, J. P. & Cole, S. W. (2015). The neuroendocrinology of social isolation. *Annual Review of Psychology, 66*, 733-767.

Clowney, E. J., Iguchi, S., Bussell, J. J., Scheer, E. & Ruta, V. (2015). Multimodal Chemosensory Circuits Controlling Male Courtship in Drosophila. *Neuron, 87*, 1036-1049.

Daly, M., Wilson, M. & Vasdev, S. (2001). Income inequality and homicide rates in Canada and the United States. *Canadian Journal of Criminology, 43*, 219-236.

DeBruine, L. M., Jones, B. C., Crawford, J. R., Welling, L. L. M. & Little, A. C. 2010 The health of a nation predicts their mate preferences: cross-cultural variation in women's preferences for masculinized male faces. *Proceedings of the Royal Society of London B, 277*, 2405–2410.

Fletcher, G. J. O., Simpson, J. A., Thomas, G. & Giles, L. (1999). Ideals in intimate relationships. *Journal of Personality and Social Psychology, 76*, 72-89.

Gangestad, S. W., Haselton, M. G. & Buss, D. M. (2006). Toward an integrative understanding of evoked and transmitted culture: the importance of specialized psychological design. *Psychological Inquiry, 17*, 138–151.

Gangestad, S. W., & Simpson, J. A. (2000). The evolution of human mating: Trade-offs and strategic pluralism. *Behavioral and Brain Sciences, 23*, 675–687.

Gonzaga, G. C., Turner, R. A., Keltner, D., Campos, B & Altemus, M. (2006). Romantic love and sexual desire in close relationships. *Emotion, 6*, 163-179.

Hassebrauck, M. & Fehr, B. (2002). Dimensions of relationship quality. *Personal Relationships, 9*, 253-270.

Hatfield, E. & Rapson, R. L. (2005). *Love and Sex: Cross-cultural perspectives*. Lanham: University Press of America.

Huston, T. L., Caughlin, J. P., Houts, R. M., Smith, S. E. & George, L. J. (2001). The connubial crucible: Newlywed years as predictors of marital delight, distress, and divorce. *Journal of Personality and Social Psychology, 80*, 237-252.

Jankowiak, W. R. (2008). *Intimacies: Love and Sex across Cultures*. New York: Columbia University Press.

Jankowiak, W. R. (1995). *Romantic Passion: A Universal Experience?* New York: Columbia University Press.

Jankowiak, W. R., Volsche, S. L. & Garcia, J. R. (2015). Is the Romantic-Sexual Kiss a Near Human Universal? *American Anthropologist, 117*, 535-539.

Jankowiak, W. R. & Paladino, T. (2008). Desiring sex, longing for love: A tripartite conundrum. In *Intimacies: Love and Sex across Cultures*, Ed. W. R. Jankowiak. New York: Columbia University Press.

Jankowiak, W. R. & Fischer, E. F. (1992). A cross-cultural perspective on romantic love. *Ethnology, 31*, 149-155.

Kandrik, M., Jones, B. C. & DeBruine, L. M. (2015). Scarcity of female mates predicts regional variation in men's and women's sociosexual orientation across US states. *Evolution and Human Behavior, 36*, 206-210.

Little, A. C., Jones, B. C. & DeBruine, L. M. (2011a). Facial attractiveness: Evolutionary based research. *Philosophical Transactions of the Royal Society B, 366*, 1638-1659.

Marston, P. J., Hecht, M. L., Manke, M. L., Mcdaniel, S. & Reeder, H. (1998). The subjective experience of intimacy, passion, and commitment in heterosexual loving relationships. *Personal Relationships, 5*, 15-30.

Miller, S. L. & Maner, J. K. (2012). Overperceiving disease cues: The basic cognition of the behavioral immune system. *Journal of Personality and Social Psychology, 102*, 1198-1213.

Murray, D. R. & Schaller, M. (2010). Historical prevalence of infectious disease within 230 geopolitical regions: A tool for investigating origins of culture. *Journal of Cross Cultural Psychology, 41*, 99–108.

Pollet, T. V., Stoevenbelt, A. H. & Kuppens, T. (2017). The potential pitfalls of studying adult sex ratios at aggregate levels in humans. *Philosophical Transactions of the Royal Society of London B, 372*, 20160317.

Posse, J. L., Dios, P. D. & Scully, C. (2017). Infection Transmission by Saliva and the Paradoxical Protective Role of Saliva. *Saliva Protection and Transmissible Diseases*, 1-18.

Thistle, R., Cameron, P., Ghorayshi, A., Dennison, L. & Scott, K. (2012). Contact Chemoreceptors Mediate Male-Male Repulsion and Male-Female Attraction during Drosophila Courtship. *Cell, 149*, 1140-1151.

Todd, P. M., Penke, L., Fasolo, B. & Lenton, A. P. (2007). Different cognitive processes underlie human mate choices and mate preferences. *Proceedings of the National Academy of Sciences USA, 104*, 15011-15016.

Trivers, R. L. (1972). Parental investment and sexual selection. In B. Campbell (Ed.), *Sexual selection and the descent of man*, 1871–1971 (pp. 136–179). London: Heinemann.

Tybur, J. M., Inbar, Y., Aarøe, L., Barclay, P., Barlow, F. K., de Barra, M., et al. (2016). Parasite stress and pathogen avoidance relate to distinct dimensions of political ideology across 30 nations. *Proceedings of the National Academy of Sciences USA, 113*, 12408-12413.

Tybur, J. M. & Lieberman, D. (2016). Human pathogen avoidance adaptations. *Current Opinion in Psychology, 7*, 6-11.

Wlodarski, R. & Dunbar, R. I. M. (2014). What's in a Kiss? The Effect of Romantic Kissing on Mate Desirability. *Evolutionary Psychology, 12*, 178-199.

Wlodarski, R. & Dunbar, R. I. (2013a). Examining the possible functions of kissing in romantic relationships. *Archives of Sexual Behaviour, 42*, 1415-1423.

Wlodarski, R. & Dunbar, R. I. (2013b). Menstrual cycle effects on attitudes toward romantic kissing. *Human Nature, 24*, 402-413.